

Math 31B — Final Review: one map for the whole course | the answer to each example is given — work it yourself first, then check

the whole course on one page | black = structure | gray = when you reach for it | red = a problem you have already done | dashed red = the three links between stations

Wednesday's exam prints the trig reduction formulas, the entire convergence-test table — ten rows, statements *and* hypotheses — and five Taylor series: e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $\arctan x$. $\sum_{n \geq 0} x^n = \frac{1}{1-x}$ is *not* one of those five; you read it off the geometric row of the test table instead. What the sheet never gives you is which tool to reach for — and that is what this page is for.

1 $\int f(x) dx$ — “what is it?”

u-substitution you can see g and g' sitting in the integrand

ex $\int_1^2 \frac{x^2}{x^3+4} dx$ (ET2) $= \frac{1}{3} \ln \frac{12}{5}$

Integration by parts two unrelated factors; or $\ln x$ / $\tan^{-1} x$ standing alone

ex $\int_0^1 x e^{-2x} dx$ (ET3) $= \frac{1-3e^{-2}}{4}$

Trig integrals powers of $\sin$, $\cos$, $\tan$, $\sec$ — reduction formulas are on the exam

ex $\int \cos(7\theta) \cos^2(7\theta) d\theta$ (WS3 1e) $= \frac{1}{7} \sin 7\theta - \frac{1}{21} \sin^3 7\theta + C$

Trig substitution $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$

ex $\int_{-3/2}^{3/2} \sqrt{9-4x^2} dx$ (MT1 #9) $= \frac{9\pi}{4}$

Partial fractions a rational function whose denominator factors

ex $\int \frac{dx}{(x^2+10)(x-4)^2}$ (MT1 #2, *form* only) $= \frac{A}{x-4} + \frac{B}{(x-4)^2} + \frac{Cx+D}{x^2+10}$

FTC part 1 + chain rule $\frac{d}{dx}$ of an integral with a variable limit

ex $\frac{d}{dx} \int_0^{\ln x} \sqrt{1+t^4} dt$ (MT1 #6) $= \frac{\sqrt{1+(\ln x)^4}}{x}$

Area / net change “area bounded by”, “net change” — split where the curves cross

ex area between x^3 and x on $[-1, 1]$ (MT1 #10) $= \frac{1}{2}$

2 Improper $\int$ — “does it exist?”

Rewrite as a limit — always line one ①

$\int_1^\infty = \lim_{b \rightarrow \infty} \int_1^b$, and split at every bad point

ex $\int_{-\infty}^0 e^x dx$ (ET5) $= 1$; the limit notation *is* the five points

p-test at ∞ : converges iff $p > 1$; at 0: iff $p < 1$.

Opposite directions — check which end is bad.

ex $\int_0^1 \frac{dx}{x^5}$ (MT2 1c) diverges

Comparison bound *above* to prove convergence, *below* to prove divergence

ex $\int_1^\infty \frac{dx}{x^2 + \sin^2 x}$ (WS5 #5) $\leq \int_1^\infty \frac{dx}{x^2}$: converges

WS5 #7 (Herbert) bounds above and then **wrongly** concludes divergence: an upper bound can never prove divergence. The direction is the entire problem.

3 $\{a_n\}$ — “does $a_n \rightarrow L$?”

Pass to $f(x)$, then L'Hôpital a sequence has no derivative: move to f with $a_n = f(n)$, take the limit, come back. One-way: f having no limit settles nothing about a_n .
ex $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$ (ET6) $\rightarrow 0$

Squeeze a $\sin$, $\cos$ or $n!$ you cannot evaluate, trapped between two limits you can

ex $1 \leq a_n \leq 2^{1/n}$ (MT2 #9) $\rightarrow 1$

ex $\frac{\cos n}{n!}$ (WS6 #6) $\rightarrow 0$

The sign: $|a_n|$ is not a_n ② $|a_n| \rightarrow L$ does *not* give $a_n \rightarrow L$ — *except* when $L = 0$

ex $\cos(\pi n)$ (WS6 2d): $= (-1)^n$, so $|a_n| \rightarrow 1$ but the limit does not exist

Geometric ratio $r^n \rightarrow 0$ exactly when $|r| < 1$

ex $\frac{3^{n+2}}{5^n}$ (MT2 8b) $\rightarrow 0$

4 $\sum a_n$ — “does S_N converge?” (and only sometimes: “what is the sum?”)

A. You can produce the sum.

Geometric cr^n with $c \neq 0$: converges exactly when $|r| < 1$, to $\frac{\text{first term}}{1-r}$ — read the first term off the term you actually have, not off $n = 0$

ex $\sum_{n \geq 1} (2/3)^n$ (WS7 1a) $= 2$; $\sum_{n \geq 0} \frac{3^{n+1} - 2(-1)^n}{2^{2n}}$ (MT2 #10) $= \frac{52}{5}$

Telescoping / partial sums consecutive terms cancel — write S_N and let $N \rightarrow \infty$

ex $\sum_{n \geq 7} (\frac{1}{n} - \frac{1}{n+1})$ (WS7 1c) $= \frac{1}{7}$; $\sum_{n \geq 1} (\sqrt{n+1} - \sqrt{n})$ (ET7): $S_N = \sqrt{N+1} - 1 \rightarrow \infty$, diverges *although* $a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0$

B. You can only answer yes or no.

Divergence Test — run it first, every time ② if $a_n \not\rightarrow 0$ it diverges; if $a_n \rightarrow 0$ you have learned nothing and must keep going

ex $\sum (-1)^{n+1} \frac{n}{n+1}$ (WS8 #2): a_n has no limit, so the series diverges

p-series, and the Integral Test ① for the Integral Test say *positive, continuous, decreasing* out loud — a graded step

ex $\sum_{n \geq 2} \frac{1}{n(\ln n)^2}$ (WS7 2b): converges, since $\int_2^\infty \frac{dx}{x(\ln x)^2} = \frac{1}{\ln 2}$ is finite

Direct and Limit Comparison the terms *look like* a p -series or a geometric once you keep only the largest piece. Both need positive terms; for LCT also say $\lim a_n/b_n$ is finite and positive.

ex $\sum \frac{2+\sin n}{n\sqrt{n+\ln n}}$ (MT2 12a): $\leq \sum \frac{3}{n^{3/2}}$, converges; $\sum \frac{n}{n^2+1}$ (WS7 2e): against $\frac{1}{n}$, diverges

Alternating Series Test — but step one is always $\sum |a_n|$ if it converges you are done and the answer is *absolute*; only when it fails do you reach for the AST

ex $\sum_{n \geq 1} \frac{(-1)^{n+8}}{n^2+4}$ (ET8): $\frac{8}{n^2+4} < \frac{8}{n^2}$, so **absolute**; $\sum_{n \geq 2} \frac{(-1)^n}{n \ln n}$ (WS8 3a): step one fails, AST works, so **conditional**

Ratio and Root factorials, n th powers, c^n . The statement is printed; *when* to reach for it is not. $L = 1$ says nothing; $L = 0$ is a fine $L < 1$.

ex $\sum \frac{3^n}{(n+1)!}$ (ET9): ratio $= \frac{3}{n+2} \rightarrow 0$, converges; $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ (WS9 #1): the same $L = 1$, opposite verdicts

5 $\sum a_n(x-c)^n$ — “for which x ?”

Interval of convergence Ratio Test on the *terms* gives R . R is not the answer — then test each endpoint by hand. Putting $x = c \pm R$ into the whole term $a_n(x-c)^n$ turns the power series into an ordinary number series; test *that*, never $\sum a_n$ on its own.

ex $\sum_{n \geq 1} \frac{3^n(x-2)^n}{n4^n}$ (WS9 5a): $R = \frac{4}{3}$, interval $[\frac{2}{3}, \frac{10}{3})$ — at the left end $\sum_{n \geq 1} \frac{(-1)^n}{n}$ converges, at the right end $\sum_{n \geq 1} \frac{1}{n}$ diverges

ex $\sum_{n \geq 1} \frac{n!}{5^n} (x-3)^n$ (WS9 5b): $R = 0$, so it converges only at $x = 3$ and there are no endpoints to test

Build it from what you are given ③ five Taylor series are printed. $\frac{1}{1-x} = \sum_{n \geq 0} x^n$ for $|x| < 1$ is not among them — it is the geometric row of the test table, and it is what WS9 #4, WS10 #2 and ET10 all run on. Four moves: substitute, multiply, differentiate, integrate.

ex $\frac{1}{x}$ at $c = 1$ (ET10): write $x = 1 + (x-1)$; the denominator is then $1-r$ with $r = -(x-1)$, so $\frac{1}{x} = \sum_{n \geq 0} (-1)^n (x-1)^n$ on $|x-1| < 1$

ex $(x^2+2x)e^{3x}$ at $c = 0$ (Lec 13 HW 5d): substitute $3x$ into e^x , then multiply: $\sum_{n \geq 0} \frac{3^n}{n!} (x^{n+2} + 2x^{n+1})$, all x

Coefficient $\leftrightarrow$ derivative $a_n = \frac{f^{(n)}(c)}{n!}$, so $f^{(n)}(c) = n! a_n$. A coefficient is a number, not a derivative.

ex from $\sum \frac{(n+1)x^n}{3^n}$ (Lec 13 HW 4): $f^{(99)}(0) = 99! \cdot \frac{100}{3^{99}}$

① **2 $\leftrightarrow$ 4** the one place an improper integral answers a series question: $\sum_{n \geq 2} \frac{1}{n(\ln n)^2}$ becomes $\int_2^\infty \frac{dx}{x(\ln x)^2}$. ② **3 $\leftrightarrow$ 4** $\sum a_n$ converges *means* $\{S_N\}$ converges, so station 3's tools are available for S_N ; and the Divergence Test is a station-3 limit settling a station-4 question. ③ **1 $\leftrightarrow$ 5** differentiate or integrate a power series termwise on $|x| < R$: $\frac{1}{1-x^4} \mapsto \frac{4x^3}{(1-x^4)^2}$ and $\frac{1}{1+2x} \mapsto \frac{\ln(1+2x)}{2}$ (Lec 13 HW 1, 2). The radius holds, but an endpoint can move — the second gains $x = \frac{1}{2}$.

Math 31B — Final Review: writing it down

The front picks the tool and gives you the answer. This side is the part that is actually graded — five of those same examples, written out the way the exam wants them.

The formula sheet prints the tests together with their hypotheses. That is not so you can skip them — it is so you can copy them into your proof. “positive, continuous, decreasing” and “ b_n positive, decreasing, $\rightarrow 0$ ” are marks you get only by writing them on the page. Everything below uses only tests whose statements are printed on your exam.

ET5 and MT2 1(c) — two bad points

Evaluate $\int_{-\infty}^0 e^x dx$.

The integral is improper at the lower limit, so *by definition*

$$\int_{-\infty}^0 e^x dx = \lim_{a \rightarrow -\infty} \int_a^0 e^x dx.$$

For each finite a , $\int_a^0 e^x dx = [e^x]_a^0 = 1 - e^a$. Since $e^a \rightarrow 0$ as $a \rightarrow -\infty$, that limit is $1 - 0 = 1$.

The limit exists and is finite, so the integral **converges, with value 1**.

A one-line antiderivative, and still four lines of writing. The definition line and the closing sentence each carry marks of their own.

Does $\int_0^1 \frac{dx}{x^5}$ **converge?**

Here the integrand blows up at $x = 0$, the *lower* limit, so

$$\int_0^1 \frac{dx}{x^5} = \lim_{a \rightarrow 0^+} \int_a^1 x^{-5} dx.$$

For each $a > 0$, $\int_a^1 x^{-5} dx = \left[\frac{x^{-4}}{-4} \right]_a^1 = -\frac{1}{4} + \frac{1}{4a^4}$, and this $\rightarrow \infty$ as $a \rightarrow 0^+$, so the integral **diverges**.

Or cite the p -test at 0: $\int_0^1 dx/x^p$ converges only when $p < 1$, and here $p = 5$. Watch the direction — at ∞ it is the opposite way round.

ET8 and WS8 3(a) — two steps, opposite verdicts

ET8. $\sum_{n=1}^{\infty} \frac{(-1)^n 8}{n^2 + 4}$: absolutely, conditionally, or divergent?

Step 1 — test $\sum |a_n|$ first.

$\sum |a_n| = \sum \frac{8}{n^2 + 4}$. For every $n \geq 1$, $0 < \frac{8}{n^2 + 4} < \frac{8}{n^2}$, and $\sum \frac{8}{n^2}$ converges (it is 8 times a p -series with $p = 2 > 1$). By the **Direct Comparison Test**, $\sum |a_n|$ converges.

Step 1 succeeded, so the series **converges absolutely**. There is no Step 2 — the Alternating Series Test is never used.

WS8 3(a). $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$.

Step 1 — test $\sum |a_n|$ first.

$\sum |a_n| = \sum_{n \geq 2} \frac{1}{n \ln n}$. Let $f(x) = \frac{1}{x \ln x}$. On $[2, \infty)$ this f is *positive, continuous and decreasing*, so the Integral Test applies:

$$\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} \int_2^b \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} [\ln(\ln x)]_2^b = \infty.$$

By the **Integral Test**, $\sum |a_n|$ diverges. Step 1 failed.

Step 2 — only now the Alternating Series Test.

$b_n = \frac{1}{n \ln n}$ is positive, is decreasing, and $b_n \rightarrow 0$, so $\sum a_n$ converges. It converges but not absolutely: **converges conditionally**.

The two proofs differ in exactly one place — whether Step 1 succeeded.

“Conditionally” is a word you are allowed to write only after Step 1 has failed.

WS9 5(a) — an interval of convergence

Find every x for which $\sum_{n=1}^{\infty} \frac{3^n(x-2)^n}{n 4^n}$ converges.

Write $t_n = \frac{3^n(x-2)^n}{n 4^n}$. For $x \neq 2$ every $t_n \neq 0$, so the Ratio Test applies to the *terms*:

$$\left| \frac{t_{n+1}}{t_n} \right| = \frac{3|x-2|}{4} \cdot \frac{n}{n+1} \rightarrow \frac{3}{4}|x-2| = L.$$

$L < 1$ exactly when $|x-2| < \frac{4}{3}$, so $R = \frac{4}{3}$ about the center $c = 2$, and the endpoints are $2 \pm \frac{4}{3}$.

$x = \frac{2}{3}$:

$t_n = \frac{3^n(-4/3)^n}{n 4^n} = \frac{(-1)^n}{n}$. Here $b_n = \frac{1}{n}$ is positive, decreasing, and $\rightarrow 0$, so the **Alternating Series Test** gives convergence. **Include $\frac{2}{3}$** .

$x = \frac{10}{3}$:

$t_n = \frac{3^n(4/3)^n}{n 4^n} = \frac{1}{n}$, a p -series with $p = 1$, which diverges. **Exclude $\frac{10}{3}$** .

Answer: $\left[\frac{2}{3}, \frac{10}{3} \right)$.

R is not the answer. Substituting an endpoint turns the power series back into an ordinary number series, and the two endpoints are two separate problems — here they disagree.

What a full-credit proof contains

- Name the test**, in words.
- Check its hypotheses out loud**. They are printed on your exam, so copying them onto your page costs nothing and is where the marks are.
- Write the limit as a limit**. $\lim_{b \rightarrow \infty}$, $\lim_{n \rightarrow \infty}$ — on the page, not in your head.
- Compare your result to the number the test uses**: to 1 for the Ratio and Root Tests, to 0 for the Divergence Test. For a p -series the question is whether $p > 1$.
- End with a sentence** carrying both the verdict and the name of the test.

Six ways to lose marks while writing correct mathematics

- “Conditionally” after proving only convergence**. The Alternating Series Test says the series converges; it cannot say which kind, because it never looks at $\sum |a_n|$. *Instead*: make $\sum |a_n|$ a numbered Step 1, every time.
- Computing the ratio but never taking the limit**. “ $\frac{3}{n+2} < 1$, so it converges” is not the Ratio Test. The test is about $L = \lim$, and a ratio below 1 for every single n can still give $L = 1$ — $\sum \frac{1}{n}$ does exactly that. *Instead*: take the limit, call it L , then compare L to 1.
- Reading $L = 0$ as inconclusive**. $0 < 1$. Only $L = 1$ is inconclusive.
- Stopping at R** . R answers a smaller question than the one asked. *Instead*: substitute each endpoint and test the two number series separately — they can disagree.
- Comparing in the wrong direction**. An upper bound can never prove divergence and a lower bound can never prove convergence. *Instead*: to prove convergence, bound above by something convergent; to prove divergence, bound below by something divergent.
- Never writing the limit for an improper integral**. $\int_1^{\infty} f$ is defined as $\lim_{b \rightarrow \infty} \int_1^b f$. Antidifferentiating and then substituting ∞ skips the definition the problem is testing. *Instead*: limit line first, evaluate second.